

CLASS X (CHEMISTRY COMPONENT)

PRACTICE TEST II (ANSWER KEY)

2025-26

17. When hydrated ferrous sulphate crystals are heated: (1)

- (a) a colourless, odourless gas evolved leaving behind a yellow residue.
- (b) a brown, odourless gas evolved, leaving behind a reddish-brown residue.
- (c) two colourless gases evolved, leaving behind a reddish-brown residue.
- (d) one colourless and one brown gas evolved, leaving behind a yellow residue.

Ans: (c) two colourless gases evolved, leaving behind a reddish-brown residue.

18. A metal 'X' on treatment with sodium hydroxide liberates gas 'P'. It also liberates the same gas 'P' on treatment with dilute sulphuric acid.

Based on the above information, 'X' and 'P' respectively are: (1)

- (a) Copper and sulphur dioxide
- (b) Zinc and sulphur dioxide
- (c) Copper and hydrogen
- (d) Zinc and hydrogen

Ans: (d) Zinc and hydrogen

19. The statements below show the results when three metal strips, P, Q, and R, are placed in blue copper sulphate solution.

P- Solution turns green.

Q- Solution becomes colourless.

R- Solution remains blue.

Which of the following metals could be P, Q, and R? (1)

- (a) P-Al, Q-Zn, R- Fe
- (b) P-Zn, Q-Fe, R- Ag
- (c) P-Fe, Q-Zn, R-Ag
- (d) P- Fe, Q-Al, R- Zn

Ans: (c) P-Fe, Q-Zn, R-Ag

20. Which of the following observations correctly shows the action of indicator on sodium hydroxide solution? (1)

	METHYL ORANGE	PHENOLPHTHALEIN
(a)	Orange to yellow	Remains colourless
(b)	Orange to pink	Remains colourless
(c)	Orange to yellow	Colourless to pink
(d)	Remains orange	Remains pink

Ans: (c)

21. The pH of the soil is tested, and for the better growth of crops, slightly alkaline soil is required. Which ion in the fertilizer will increase the alkalinity of the soil? (1)

- (a) Hydronium ion
- (b) Hydroxide ion
- (c) Hydrogen ion
- (d) Both hydroxyl and hydrogen

Ans:(b) Hydroxide ion

22. Which of the following substances will not give carbon dioxide on treatment with dilute acid? (1)

- (a) marble
- (b) Limestone
- (c) Baking soda
- (d) Lime

Ans: (d) Lime

23. Consider the following chemical equation:

$X(aq) + \text{sodium chloride}(aq) \rightarrow Y \text{ (white ppt)} + \text{Sodium nitrate}(aq)$ (1)

	X	Y
(a)	Silver nitrate	Silver chloride
(b)	Sodium chloride	Silver nitrate
(c)	Sodium nitrate	Silver chloride
(d)	Silver chloride	Silver nitrate

Ans: (a)

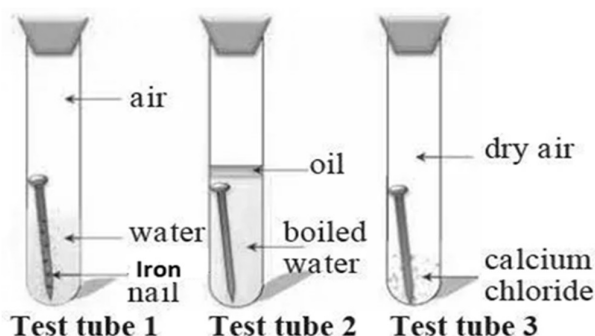
24. **Assertion (A):** Alkenes, alkynes and alkanes are examples of homologous series. (1)

Reason (R): Organic compounds of the homologous series have similar structures but different chemical properties.

Ans: A is true but R is false

25. The following activity is set up in the science lab by the teacher.

He put clean iron nails in each of the three test tubes and left them for few days. Students were asked to note down their observations. (2)



- (i) What will the students observe?
(ii) How will their observation change if copper and silver items are kept exposed in air for few days?

Ans: (i) Iron nail in test tube 1 will develop a reddish-brown coating / start rusting.

(ii) Copper will develop a green coating whereas silver develops a black coating.

26 Attempt either option A or B

(3)

A. The oxide of a metal 'T' having 3 electrons in its M-shell is amphoteric in nature.

(i) Identify the metal 'T' and explain the amphoteric nature of its oxide with the help of chemical equations.

(ii) Under what condition does the metal 'T' react with water?

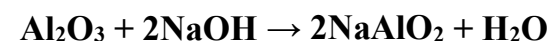
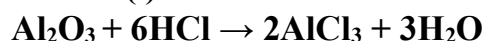
OR

B. Radhika performed an experiment in a science laboratory to study the burning of metals in air. She burnt a metal 'X' that produced a black oxide 'B' and a metal 'Y' that burned with a dazzling white flame to produce a white ash 'C'.

(i) Identify metals X and Y and state the position of X with respect to Y in the reactivity series of metals.

(ii) Draw the electron dot structure of the oxide of metal Y.

Ans: A.(i) T - Aluminium / Al



(iii) Aluminium reacts with steam.

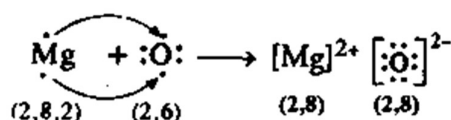
OR

B. (i) X- Copper / Cu

Y- Magnesium / Mg

X is below Y in metal reactivity series.

(ii)



27. Riya wishes to whitewash the walls of her house. She bought quick lime and mixed it with water. On mixing, she observed that the mixture started boiling even though it was not being heated. After some days she noticed a shiny finish on the walls.

(i) Identify the substance responsible for the shiny finish on the walls.

(ii) Give balanced equations for the reactions involved.

(iii) What happens when the substance identified in (i) is heated in air? Name the type of reaction that takes place. (3)

Ans: (i) Calcium carbonate / CaCO_3

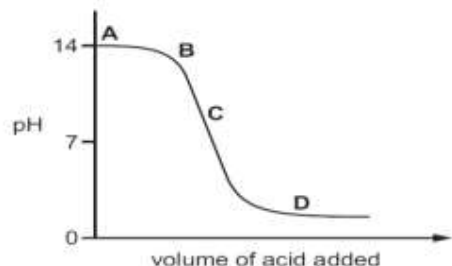
(ii) $\text{CaO} + \text{H}_2\text{O} \rightarrow \text{Ca(OH)}_2$

$\text{Ca(OH)}_2 + \text{CO}_2 \rightarrow \text{CaCO}_3 + \text{H}_2\text{O}$

(iii) It decomposes to calcium oxide and carbon dioxide gas is evolved.

Thermal decomposition reaction.

28. The graph given below depicts a neutralization reaction ($\text{acid} + \text{alkali} \rightarrow \text{salt} + \text{water}$). The pH of a solution changes as we add excess of acid to an alkali.



A. Which letter denotes the area of the graph where both acid and salt are present?

B. What will be the colour change from B to D if 1 drop of phenolphthalein is used as indicator?

OR

‘pH has great importance in our daily life’. Justify the statement by giving one example.

C. While diluting an acid, why is it recommended that the acid should be added to water and not water to acid? (4)

Ans: A. letter D

B. Pink to colourless.

OR

Example of tooth decay / use of antacids or any other example.

C. The process of dissolving an acid in water is a highly exothermic one. If water is added to a concentrated acid, the heat generated may cause the mixture to splash out and cause burns. The glass container may also break due to excessive local heating.

29. Attempt either option A or B

(5)

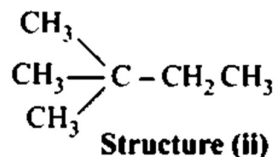
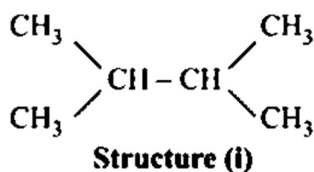
A.

A neutral organic compound 'X' (molecular formula C_2H_6O) on reacting with acidified $K_2Cr_2O_7$ gives an organic compound 'Y' which is acidic in nature. 'X' reacts with 'Y' on warming in presence of conc. H_2SO_4 to give a sweet-smelling compound 'Z'.

- Identify X, Y and Z
- Draw the electron dot structure of 'Y'.
- Name the reaction which occurs when 'Z' reacts with an alkali.
- Write a balanced chemical equation for the combustion of 'X'.

OR

B.



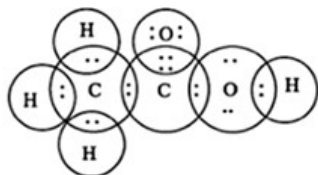
- How are structure (i) and structure (ii) given above related to each other? Give reason to justify your answer.
- Draw one more possible structure for the above case.
- Differentiate between saturated and unsaturated carbon compounds on the basis of their general formula.
- Why are carbon and its compounds used as fuel for most applications?

Ans: (a) X - Ethanol / C_2H_5OH

Y- Ethanoic acid / CH_3COOH

Z – Ethyl ethanoate / $CH_3COOC_2H_5$

(b)



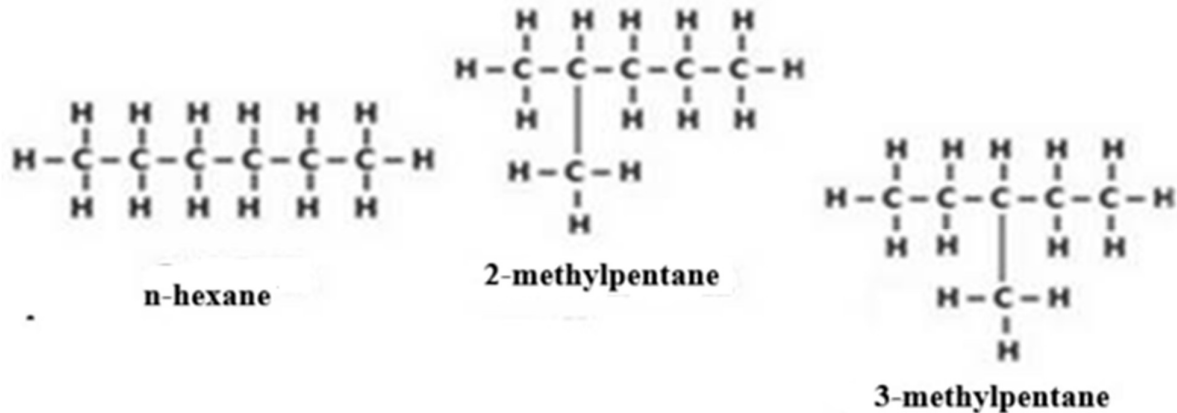
(c) Saponification

(d) $C_2H_5OH + 3O_2 \rightarrow 2CO_2 + 3H_2O$
(X)

OR

(a) Isomers of each other. Since they have same molecular formula but different structure.

(b)



(any one)

(c) Saturated compound: C_nH_{2n+2} or any other correct general formula.

Unsaturated compound: C_nH_{2n} / C_nH_{2n-2} or any other correct general formula.

(d) Carbon and its compounds undergo combustion to release large amounts of heat and light energy.